

Risk-Aware Planning for Autonomous Driving

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Area of Research: Motion Planning under Uncertainty

Autonomous driving requires a vehicle to perceive the environment, predict other road users, plan a safe trajectory, and control its motion.

This project focuses on motion planning under **uncertainty**:

- how the ego car decides a safe action **when another vehicle may slow down, continue, or move differently.**



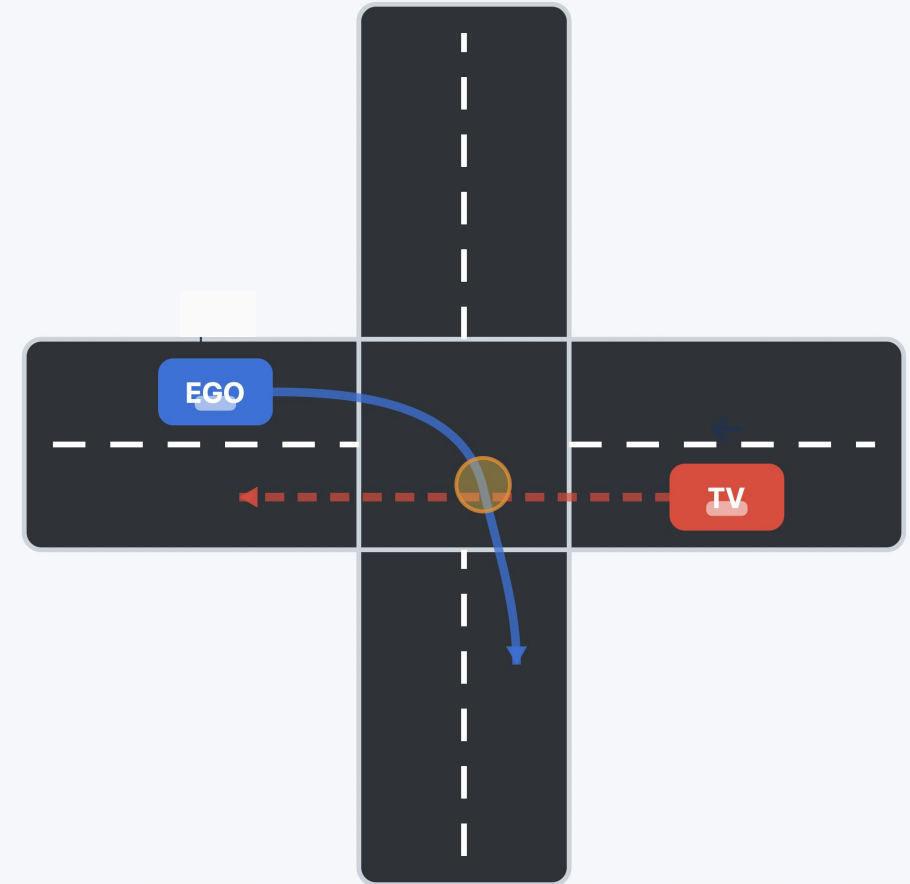
- This is not about detecting objects.
- It is about turning uncertain predictions into safe driving decisions.

Problem: Crossing an Intersection Safely

- The ego vehicle must cross an intersection while another vehicle may behave in different ways.
- The target vehicle may continue, slow down, or move slightly differently from what is expected.
- The decision must be updated repeatedly as new observations arrive.

Like crossing a road: look, think, then move.

This is a sequential decision-making problem under uncertainty.



Core Problem: One Predicted Future Is Not Enough

If the planner trusts only one guessed target-vehicle trajectory, it may:

- choose an unsafe action
- stop too conservatively
- or fail to make progress.

One guessed future



Too confident

Several possible futures



Risk-aware

The research problem is:

how can the car use multiple predicted target-vehicle trajectories to choose a safe action?

Significance: Safety Requires More Than Stopping

Intersections are high-risk scenarios because vehicles interact closely and decisions must be made quickly.

A useful planner must balance three goals:

Safety

avoid collision

Progress

avoid unnecessary
stopping or hesitation

Comfort

avoid sudden braking or
steering

- This is worth prioritising because prediction errors become real safety risks only when they affect the vehicle's action.
- If solved, the vehicle can cross complex intersections more reliably without simply becoming over-cautious.

State of the Art: From Prediction to Safe Planning

Approach	Contribution	Remaining limitation
Multimodal prediction, e.g. MultiPath	predicts several possible trajectories	does not choose the ego action
Learning-based planners	can learn complex behaviour	hard to guarantee safety and interpret failures
Rule-based safety methods	provide simple safety guarantees	may be too rigid in interactive scenarios
Risk-aware SMPC	plans under uncertainty	can become conservative or infeasible

Remaining Gap:

we need a closed-loop planner that uses multimodal predictions while balancing safety, progress, feasibility, and comfort.



SMPC = Stochastic Model Predictive Control: a planning method that chooses safe actions while accounting for uncertainty.

Hypothesis

Research question:

Can a self-driving car cross an intersection more safely by considering several possible futures?

Hypothesis:

- Using multimodal predictions in SMPC should keep the ego vehicle safer without making it unnecessarily hesitant.

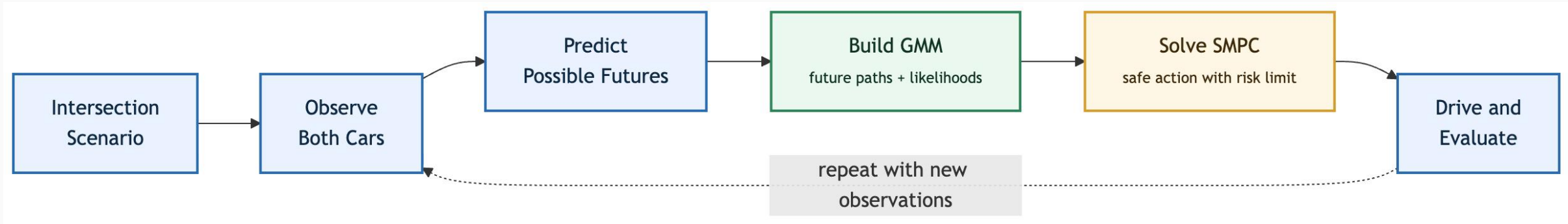
Success criteria:

- Higher minimum distance to the target vehicle.
- High solver feasibility.
- Reasonable completion time.
- Smooth and stable control actions.

Failure condition & Alternative explanation:

- If the method is unsafe, infeasible, or only stays safe by stopping too much

Execution plan



Testing: compare policies

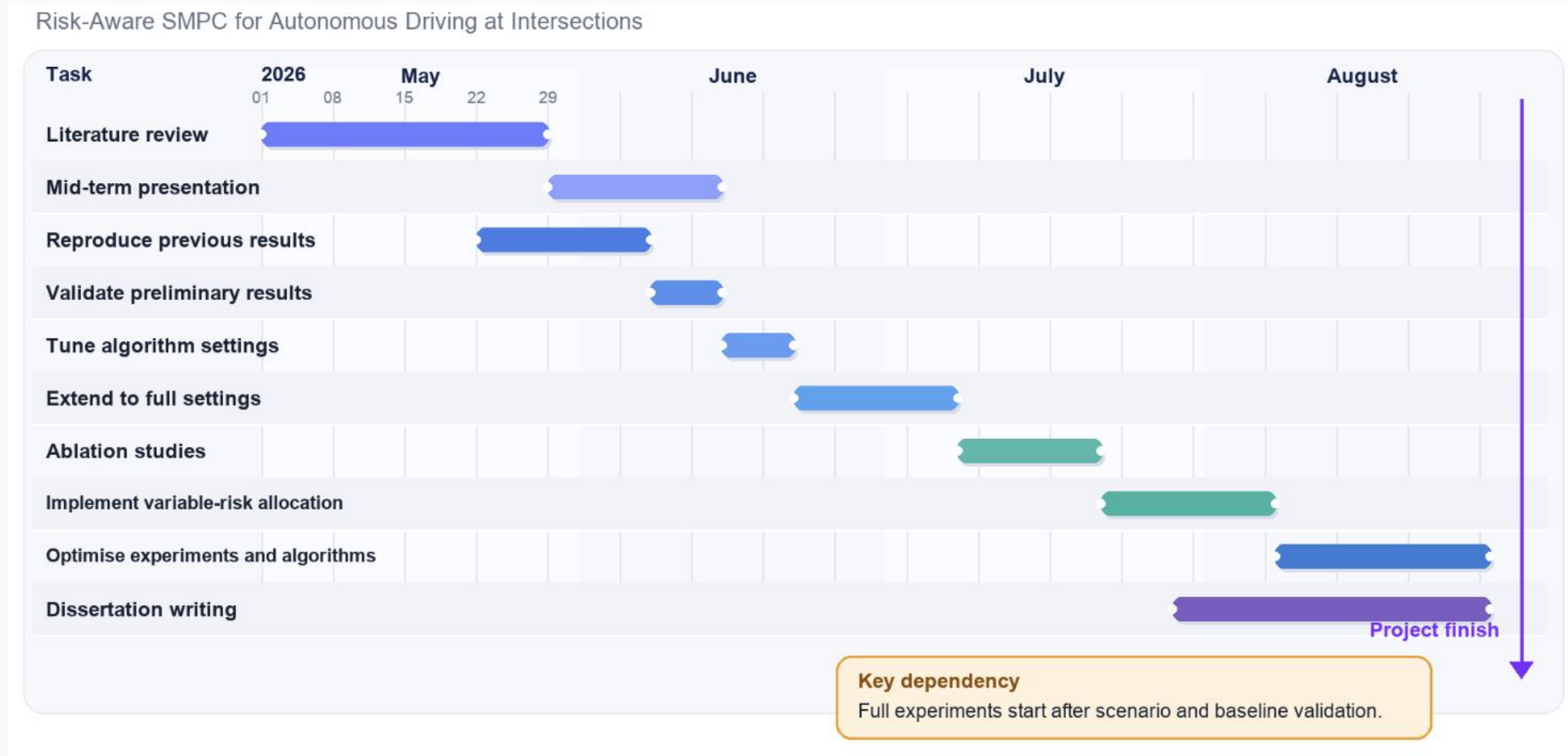
- No target-vehicle baselines.
 - **ideal path and time**
- Fixed-risk SMPC.
 - **basic uncertainty handling**
- Variable-risk SMPC.
 - **adaptive risk-budget allocation**

Evaluation: use metrics

- Completion time.
- Solver feasibility.
- Minimum distance to target vehicle.
- Path deviation.
- Comfort / jerk.
- Computation time.

GMM = Gaussian Mixture Model. It represents several possible target-vehicle trajectories and their probabilities.
SMPC = Stochastic Model Predictive Control. It chooses actions using probabilistic safety constraints.

Execution plan: Timeline and Dependencies



Risk analysis

Risk	Impact	Mitigation
Simulator is slow	Experiments take time	Run small pilots first, then scale experiments
Planner fails	Car may produce invalid actions	Save debug logs, use fallback analysis
GPUs from UCL are limited	Experiments may be delayed	Use alternative GPUs such as Google Colab
Results are mixed	Hypothesis may not be fully supported	Analyse failure cases instead of hiding them

**The main risk is interpretability:
I need to know not only whether the car succeeds, but why it succeeds or fails.**



Thank you